

Experience is the Best Teacher: Personalized Vocabulary Building Within the Context of Instagram Posts and Sentences from GPT-3

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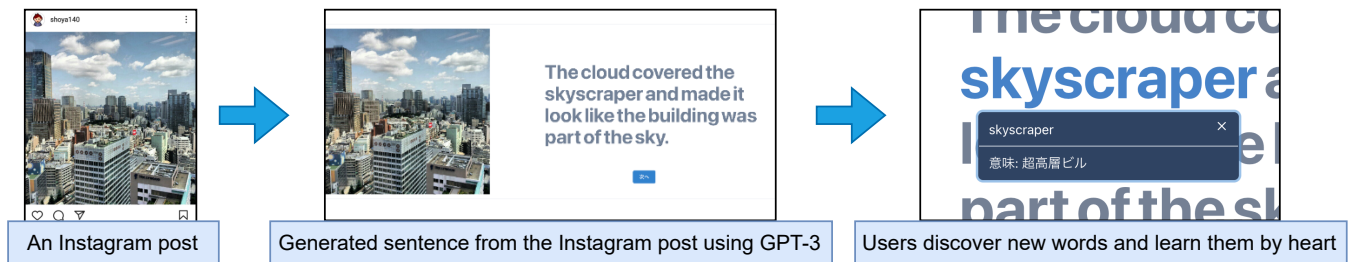


Figure 1: The proposed method to find new words within the context of Instagram posts and sentences generated from GPT-3

ABSTRACT

Although language learners have different contexts and motivations, sensing personal backgrounds to optimize learning materials has still been challenging. By focusing on the huge movement of Social Networking Services (SNS) such as Instagram, we came up with the idea of utilizing social posts, in particular images, as learning materials. This paper presents our working prototype of the proposed system that extracts keywords from these images and leverages GPT-3 to generate sentences for acquiring new vocabulary around the keywords. By conducting a pilot study involving three users, we found that 2.2 words appeared as unknown words for the user in one generated sentence on average, and there is room for improvement in the proposed system. These findings can be utilized in a large-scale evaluation designed in the future.

CCS CONCEPTS

• Applied computing → E-learning; • Human-centered computing → Social media.

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KEYWORDS

Context-Aware Language Learning; Personalized Learning; Intelligence Augmentation; GPT-3

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1 INTRODUCTION

Our experiences are considered to be stored in the brain as episodic memory. According to research in psychology and neuroscience fields, applying new information to such existing memory is more effective than doing so with non-familiar topics [1]. Associating new knowledge with existing memory with emotions, interests, or past experiences is one of the promising ways to learn things effectively while having fun.

In the past, we had few clues to determine such personal backgrounds other than asking questions in person, which is almost impossible considering the effort it requires. Today, however, social media for sharing photos like Instagram are widespread among the young generation. People describe their experiences, thoughts, or emotions on such a platform, along with pictures. These posts are informative in that people post them, especially when they are impressed, touched, or excited about something in their personal

lives. Therefore, such photos can be good sources for sensing or estimating learners' interest, his/her targets of emotions, or past experiences – to personalize content to learn and contexts where the new content can be associated to make the new content easier to remember.

This paper proposes a working-in-progress learning system to generate sentences from learners' Instagram posts (see Figure 1). A sentence generated from the system is shown on the right side of the screen. On the left side, a user's Instagram picture is shown. Our system is not intended to provide image-to-text captioning. Instead, we want to give an example sentence likely to relate to the user's experiences or interests and discover unknown but valuable vocabulary for their daily situations. By such design, users are supposed to click words in a sentence if they do not know the meanings, allowing them to show its lemma and lemma's translation in the native language of the user.

We hypothesized that associating a new vocabulary from one's personal experience – visible on his/her Instagram posts – is one of the promising ways to optimize learning materials. With this idea, we aim to enhance learning performance by finding and learning new vocabulary within the context – as a sentence and an image – that is personalized to their lives. In order to evaluate our hypothesis, we conducted a pilot study including recall tests and compared the retention ratio compared with two baselines.

2 BACKGROUND AND RELATED WORK

2.1 Context-Aware Language Learning Systems

As many researchers have reported, the context of a learner contains the potential to personalize learning materials. For instance, Jacquet *et al.* proposed *Vocabulometer*, which recommends text based on the user's vocabulary estimated from reading activity sensed by eye-tracking devices [4]. To extend their study, Yamaguchi *et al.* proposed *Mobile Vocabulometer*, where users choose topics of interest within the application, and it recommends English articles based on the selections to discover new vocabulary within the context [6]. The following topics were provided within the application, "entertainment, economy, environment, lifestyle, politics, sport, and science." While they provide preliminary work to adapt interests by user topic selection, our interest is more specific in reality. Take the topic *sport*, for example; some people are into baseball while others might like football. Our interests and experiences in daily lives are diverse; we need to sense such backgrounds, which is another motivation for our study.

Hautasaari *et al.* proposed an application called *VocaBura* which allows language learners to acquire vocabulary using their idle hours [3]. According to the study, this audio-based application allows users to find new vocabulary based on the user's location history. While they focus on personalization using location history, we focus on personalization using social image data.

2.2 Text-Generation Using GPT-3

GPT-3 is a text-to-text model where both input and output are text, allowing tasks and few-shot demonstrations by text interaction according to the original paper where GPT-3 is introduced [2]. GPT-3 is learned with 175 billion parameters, and it often provides natural responses. For example, if a text input – often called a

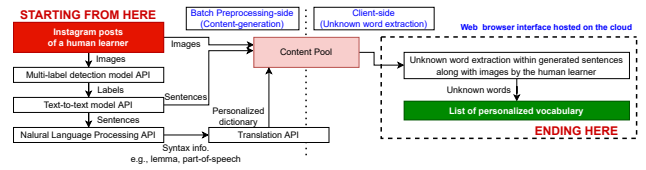


Figure 2: System overview of our proposed system.

I am a masterful journalist and can depict daily life with advanced vocabulary. I can generate an advanced sentence using some of the following keywords. I don't necessarily use all of the keywords.	Parameter	Value
Keywords: {Ten labels extracted from an image are placed here.}	model	text-davinci-002
One-Sentence: {GPT-3 completes after this.}	temperature	0.5
	max_tokens	100
	top_p	1
	frequency_penalty	0
	presence_penalty	0

(i) Our prompt to generate a sentence from keywords extracted from an image
Contents within curly braces were replaced.

(ii) Some of our parameters for GPT-3

Figure 3: Summary of GPT-3 configurations, prompt (left) and main parameters (right).

prompt – "Can I visit Japan from Germany by foot?" is given to the GPT-3, the model would respond like this: "No, you cannot visit Japan from Germany by foot. You would need to take a plane, bus, or train." While GPT-3 was announced in 2020, its API is open to everyone at the end of 2021¹. Due to its novelty, its potential use cases or usability in an educational context still has room for investigation. Therefore, our study is exploratory.

3 PROPOSED SYSTEM

As shown in Figure 2, our system comprises multiple steps. Harnessing by cloud computing APIs, our system generates labels from exported Instagram images, labels to sentences, analyzes generated sentences, and prepares translations.

3.1 Detecting Keyword-Objects from Instagram Image

Given images from learners' exported Instagram data, we feed them into a multi-label detection API – Google Cloud Vision API, providing up to ten labels per image. Take an example image from Figure 1; the following labels were detected from the image: *Cloud*, *Skyscraper*, *Building*, *Sky*, *Daytime*, *Window*, etc. We feed these labels into GPT-3 so that people can discover new words – (a) co-occurrences suggested by GPT-3 and (b) detected labels – related to the context strongly tied with the learner's experiences.

3.2 Generating Sentences by Keywords and GPT-3

We integrated GPT-3's completion API, which provides sentences to our system with our prompt and keyword labels. Currently, GPT-3 [2] is publicly accessible as a beta version. By nature of the text-to-text model, instructions to the model are given by sentences. Our prompt is described in Figure 3. Lines 1-3 clarify what the model is expected to do. Our system inserts keyword labels to the line starting from *Keywords*. GPT-3 appends a sentence after *One-Sentence*. Some work, e.g., [5], tries to find a better prompt. However, we keep our prompt simple as it is not our scope of work.

¹<https://openai.com/blog/api-no-waitlist/>

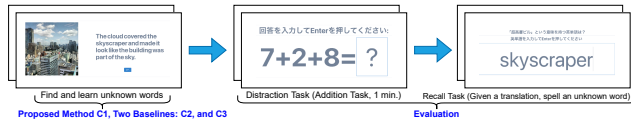


Figure 4: Overview of evaluation procedure of our system.

3.3 Providing Translations for Each Word

After all the sentences are generated, we analyze syntax by using Google Cloud Natural Language API. Through this process, the API offers morphological analysis capabilities, providing us with the following helpful information for our system: part-of-speech and lemma. We need lemmas to prevent the system from extracting multiple but essentially the same words over and over, such as *car* and *cars*. Users can click each sentence word within our system to learn them with standard forms and the corresponding translation of the standard forms.

4 EVALUATION

We conducted a pilot study shown in Figure 4 to evaluate our hypothesis: “Displaying a picture taken by the user and related to a sentence is useful in exploring unknown words for language learning.” Since this hypothesis can be decomposed into two factors (1: my image or someone’s image, 2: displaying or not displaying an image), the following three conditions were mixed on the system, and recall ratios of unknown words of each condition were compared.

- C1: Displaying an image provided by the user and a sentence generated from the image (Proposed)
- C2: Displaying an image provided by someone and a sentence generated from the image (Baseline 1)
- C3: Displaying only a sentence generated by an image provided by someone (Baseline 2)

4.1 Experimental Design

We recruited three Japanese undergraduate students who had posted at least 30 images on Instagram. This requirement of many photos was to collect data on a sufficient number of unknown words. As a pre-task of the experiment, participants submitted photos excluding sensitive data, screenshots, potential copyright issues, or texts as the main content. Pictures by others for C2 and C3 were provided by the authors. Participants who completed tasks received 1,000 JPY as a reward. This study has been approved by the ethics committee of Osaka Metropolitan University.

In the learning phase, participants clicked unknown words to check their meanings. After selecting all unknown words in one sentence, participants clicked the button to display the next sentence. This process continued until the number of unknown words reached 10 in three conditions (30 words in total) or all images prepared for C1 were shown.

As in the middle of Figure 4, the system displayed a distraction task after users extracted unknown words from the three conditions. The simple addition task of three one-digit numbers was carried out for over 1 minute. The next recall task prompted the user to

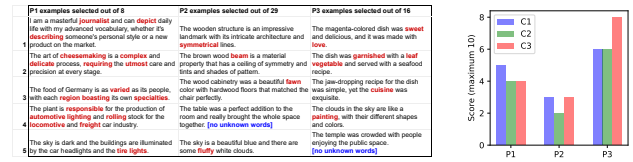


Figure 5: Results of the pilot study. Left: Random samples of generated sentences where extracted words are highlighted with red. Right: Recall task scores of all participants.

spell out English words within the user’s unknown word list in a shuffled order, providing scores for each condition.

4.2 Results

Figure 5 (left) shows examples of the sentences shown to participants during the experiment. To collect 30 unknown words (10 for each condition), 8, 29, 16 sentences were required to be displayed, respectively. To put in another form, 2.2 words appeared as unknown words in one sentence on average $((30/8 + 30/29 + 30/16)/3 = 2.2)$. We found that the ratio of words extracted per sentence varied among the users depending on their extracted words and their language skills.

We compared submitted answers and expected answers after converting them into lowercase. Scores of the recall test for each condition and participant are shown in Figure 5 (right). In P1, C1’s score outweighs other conditions. In P2, scores are the same with C1 and C3. In P3, C3 outweighs other conditions.

4.3 Discussion

Even if we had a limited number of participants, we observed findings and the following challenges that can be addressed before conducting a large-scale experiment.

Word difficulty and language skill – The number of words extracted per sentence varied depending on users. One of the possible explanations is that we focused on evaluating the proof of concept, and the system does not consider learners’ English levels. In future research, we need to make our system aware of learners’ English levels [6].

Optimization of GPT-3 parameters – In the first sentence in the case of P1 from Figure 5, we can see that our prompt, as in Figure 3, is similar to the generated sentence. This type of sentence, i.e., overfitting to line 1 of our prompt, was found only once among all the sentences, including sentences we could not show in Figure 5. Considering the amount of generated sentences shown to participants until they finished word extraction were 8, 29, and 16 (respectively for P1-P3), this unwanted behavior is considered negligible. In the future, it is better if we can prepare automatic guidelines to select proper sentences from the pedagogical point of view because GPT-3 is a novel technology.

The number of image resources – In formal English education, the levels of words are often aligned when learners before they take exams; therefore, evaluating their performance can be done by simply comparing scores within the same level, e.g., 3rd grade in high school. In our system, however, extracting words is a dynamic process. Therefore, comparing answer correctness

within similar vocabulary difficulties is crucial to evaluate our software. To do so, increasing the number of trials within an individual is essential. This, however, brings another challenge: Instagram Posts Are Finite. As a future direction, the learner's smartphone's camera rolls instead of Instagram posts. We used Instagram posts mainly because they reflect their experiences, emotion, or interests. In this sense, a tradeoff between sample size and the strength of impressions toward images is inevitable.

5 CONCLUSION

We proposed a working-in-progress learning system to generate sentences from learners' Instagram posts. We hypothesized that associating a new vocabulary with one's personal experience is one of the promising ways to optimize learning materials. We conducted a pilot study with three Japanese students. Our experiment clarified that the number of sentences they needed to extract a certain amount of unknown words varied depending on the participants. To address the issue, we need to make our system aware of learners' English levels in the future. Also, our system allows learners' to extract words, unlike formal education where word levels are aligned. Therefore, our future investigations should compare scores within similar word groups of difficulties. To do so, we need to increase posts from learners. However, since Instagram posts are finite, it is not an easy task. Therefore, using learners' camera rolls on their smartphones instead of their Instagram posts is one of our possible future research directions. While the pilot study did not provide an answer to our hypothesis, we will further extend our research from the lessons the study supplied.

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